

Exploratory Data Analysis (EDA) Findings Report

1. Introduction

This report summarizes the key findings from the Exploratory Data Analysis (EDA) performed on the Diwali Sales Dataset. The goal of the analysis was to understand data distribution, relationships, trends, and potential insights to support decision making.

2. Dataset Overview

- Description of dataset: (Replace with your dataset details)
- Number of rows: 11251
- Number of columns: 15
- Key features studied: 5

3. Data Cleaning Summary

- Missing values handled
- Outliers detected and treated
- Data type corrections
- Duplicate records removed (if any)

4. Key EDA Findings

- **Distribution Analysis:**
 - Summary statistics reveal the central tendencies and spread of data.
 - Numerical columns show patterns such as skewness or normality.
- **Correlation Insights:**
 - Strong positive/negative correlations identified among major variables.
 - Potential predictors recognized based on correlation strength.
- **Category-wise Patterns:**
 - Categorical variables show significant variation across groups/categories.
- **Trend Analysis:**
 - Time-based patterns observed (if time-series data).
- **Outliers:**
 - Outliers found is not Found in any column.
 - Their impact was analyzed and treated appropriately.

5. Visual Insights

The EDA included visualizations such as:

- Countplot for identifying each category count.
- Boxplots for identifying outliers
- Heatmap of correlations
- Bar charts for categorical insights

Each visualization helped uncover important relationships and patterns within the data.

6. Conclusion

The EDA process provided a comprehensive understanding of the dataset. It revealed data quality issues, key trends, correlations, and actionable insights that can guide further modeling or business decisions.

7. Key Findings:

- Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy an products from Food, Clothing and Electronics category.
- Most Purchased Product are (e.g- Food, Clothes, Electronics).
- Most of the buyers are Working in IT, Healthcare and Aviation Sector.
- Most of the buyers are Married(women).
- Most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively.